Rtl4 (Sirh11/Zcchc16): A Secreted Microglial Effector of the Brain Noradrenergic System

Gene: Rtl4 (Retrotransposon Gag-like protein 4; synonyms Mar4/Mart4, Sirh11, Zcchc16, Sushi-XF2b) **UniProt:** Q3URY0 (RTL4_MOUSE) **Organism:** *Mus musculus* (mouse) **Key domain:** Retroviral-type CCHC zinc finger (InterPro IPR001878 / IPR036875) **Date:** 2026-10-24

Summary

Mouse **Rtl4** — also known as **Sirh11**, **Zcchc16**, or **Mart4** — is a **eutherian-specific, X-linked gene domesticated from the Gag region of a sushi-ichi (Ty3/Gypsy) LTR retrotransposon**. It encodes a ~304-amino-acid (34.3 kDa) protein whose only recognizable structural feature is a C-terminal **retroviral-type CCHC zinc-finger (zinc knuckle)** flanked by clusters of basic residues — the signature nucleocapsid motif of retroviruses and retrotransposons. On this basis, Rtl4 is predicted by homology to be a single-stranded RNA / nucleic-acid-binding protein, although no direct biochemical assay of its activity has yet been published. The gene has been verified as the correct target: the symbol "Rtl4," the mouse organism, and the CCHC zinc-finger domain all align precisely with the literature on Sirh11/Zcchc16, and the identification is unambiguous.

The **primary, experimentally supported function of Rtl4 is in the brain**, where it acts as a **secreted microglial protein coupled to the noradrenergic (locus coeruleus) system**. Knockout mice display cognitive deficits — impaired attention, elevated impulsivity, and impaired short-term working memory — together with reduced recovery of noradrenaline (relative to dopamine) in the prefrontal cortex, directly implicating the brain noradrenergic system. Knock-in reporter mice (Rtl4–Venus fusion) subsequently revealed that Rtl4 is expressed specifically in **microglia** (not astrocytes) and that the protein is **secreted into the extracellular space** of the postnatal midbrain, hypothalamus, hippocampus, and amygdala — regions forming new neuronal networks. Critically, this secretion is **dynamically regulated by**

noradrenergic/β-adrenergic signaling and arousal state: a β-adrenergic agonist increases extracellular RTL4 signal, anesthesia decreases it, and a noradrenaline-reuptake inhibitor maintains it.

Rtl4 therefore sits at the intersection of **retrotransposon domestication, microglial biology, and monoaminergic neuromodulation**. It belongs to a small family of secreted microglial RTL/SIRH proteins; whereas its paralogs RTL5, RTL6, and RTL9 act as extracellular innate-immune pattern-capture factors against viral, bacterial, and fungal PAMPs respectively, Rtl4/Sirh11 is distinguished as the family member that **"responds to noradrenaline."** It has also been independently identified as a candidate causative gene in **autism spectrum disorder (ASD)**. The most parsimonious mechanistic model is that Rtl4 is a secreted microglial effector that participates in extracellular noradrenaline handling or DAMP/molecular scavenging during the postnatal assembly of noradrenergic networks, thereby supporting attention, impulse control, and working memory.

Key Findings

Finding 1 — Rtl4 is a domesticated sushi-ichi retrotransposon Gag gene encoding a CCHC zinc-finger protein

Rtl4 (Mart4/Sirh11/Zcchc16; UniProt Q3URY0) is **one of 11 eutherian-specific retrotransposon Gag-like (RTL) / sushi-ichi retrotransposon homolog (SIRH) genes** derived from a Ty3/Gypsy sushi-ichi LTR retrotransposon. It is a "domesticated" (exapted) gene: an ancient mobile-element Gag open reading frame that has been co-opted into the host genome and maintained as a functional cellular gene. The encoded protein exhibits high sequence identity to retrotransposon Gag and carries a **CCHC-type zinc finger**, the hallmark of retroviral/retrotransposon nucleocapsid proteins.

Evolutionary analysis supports that Rtl4 acts as a **protein**, not merely a non-coding RNA: pairwise **dN/dS (non-synonymous/synonymous substitution ratio)** comparisons between orthologs indicate the open reading frame is under purifying/functional selection consistent with protein-coding constraint. As summarized by Irie, Kaneko-Ishino, Ishino and colleagues:

> *"Sirh11/Zcchc16 encodes a CCHC type of zinc-finger protein that exhibits high homology to an LTR retrotransposon Gag protein."* — PMID: 26402067

> *"Amongst the 11 eutherian-specific genes acquired from a sushi-ichi retrotransposon is the CCHC type zinc-finger protein-encoding gene SIRH11/ZCCHC16."* — PMID: 27446905
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> *"dN/dS ratios in pairwise comparisons between its orthologs have provided supportive evidence that it acts as a protein."* — PMID: 27446905

This finding anchors the molecular identity of the gene and distinguishes Rtl4 from better-known SIRH family members such as Peg10 and Peg11/Rtl1, which function in the placenta.

Finding 2 — Knockout mice show cognitive deficits linked to the brain noradrenergic system

Targeted deletion of mouse Sirh11/Zcchc16 (Rtl4) produces **abnormal behaviors related to cognition — specifically deficits in attention, elevated impulsivity, and impaired working memory** (short-term spatial memory), along with a failure to adapt properly to novel environments. These behavioral phenotypes point to dysfunction of attentional and executive circuits.

The mechanistic link to neurochemistry came from **in vivo microdialysis of the prefrontal cortex**. After perfusion with high-potassium artificial cerebrospinal fluid (a depolarizing stimulus that evokes neurotransmitter release), the **recovery rate of noradrenaline (NA) was reduced relative to dopamine (DA)** in knockout versus control mice. Because noradrenergic projections to the prefrontal cortex arise from the locus coeruleus, this selective NA (vs DA) deficit implicates the **brain noradrenergic system**. Importantly, this made Sirh11/Zcchc16 the **first SIRH/RTL gene demonstrated to function in the brain rather than the placenta**.

> *"Gene targeting of mouse Sushi-ichi-related retrotransposon homologue 11/Zinc finger CCHC domain-containing 16 (Sirh11/Zcchc16) causes abnormal behaviors related to cognition, including attention, impulsivity and working memory."* — PMID: 26402067
>
> *"the recovery rate of noradrenaline (NA) was reduced compared with dopamine (DA) after perfusion of high potassium-containing artificial cerebrospinal fluid in knockout (KO) mice."* — PMID: 26402067
>
> *"Sirh11/Zcchc16 is the first SIRH gene to be involved in brain function, instead of just the placenta."* — PMID: 26402067

Finding 3 — Inferred molecular function: RNA/nucleic-acid binding via a retroviral CCHC zinc knuckle

The **direct biochemical activity of Rtl4 has not been assayed**, but its only recognizable domain is a retroviral-type CCHC zinc-finger (InterPro **IPR001878** Znf_CCHC; **IPR036875** Znf_CCHC_sf) — the signature motif of retroviral and retrotransposon **nucleocapsid (NC / Gag)** proteins. In the well-characterized yeast Ty1 retrotransposon, the C-terminal Gag region bearing this motif functions as a **nucleic-acid chaperone** that promotes tRNA primer annealing, RNA dimerization, and reverse-transcription initiation. By homology, Rtl4 is predicted to be a **single-stranded RNA / nucleic-acid-binding protein**.

> *"Retroviral NC proteins are formed of either one or two CCHC zinc finger(s) flanked by basic residues and have nucleic acid chaperone and match-maker properties essential for virus replication."* — PMID: 10766747

This domain-based inference is the strongest available statement about Rtl4's intrinsic molecular activity. It does not necessarily mean Rtl4 retains an intracellular RNA-packaging role; rather, the ancestral RNA-binding chemistry may have been repurposed in a secreted, extracellular context (see Mechanistic Model).

Finding 4 — A eutherian-specific, X-linked gene with lineage-specific structural variation, implicated in brain evolution

SIRH11/ZCCHC16 is an **X-linked** gene conserved in three of the four major eutherian clades — **Euarchontoglires, Laurasiatheria, and Afrotheria** — but heavily mutated / pseudogenized in **Xenarthra** (sloth, armadillo). Across 95 eutherian species examined, the open reading frame shows large **lineage-specific structural variation**, including loss of the C-terminal CCHC RNA-binding domain (gibbons, megabats) or loss of the N-terminal half (New World monkeys, some hystricognath rodents, cetaceans, ruminants), and apparent loss of normal function in most gibbon genera. Because deletion of the gene impairs cognition in mice, its original acquisition is proposed to have contributed to **eutherian brain evolution**, with subsequent lineage-specific modifications possibly underlying species-specific neurological differences.

> *"SIRH11/ZCCHC16 is an X-linked gene that encodes a CCHC type of zinc-finger protein that exhibits high sequence identity to the LTR retrotransposon Gag protein."* — PMID: 28484174

> *"It became a pseudogene in armadillos (Cingulata) and sloths (Pilosa), the only two extant orders of xenarthra."* — PMID: 27446905
>
> *"it is highly conserved in three out of the four major groups of the eutherians, euarchontoglires, laurasiatheria and afrotheria, but is heavily mutated in xenarthran species such as the sloth and armadillo."* — PMID: 26402067

Finding 5 — RTL4 is a microglial gene whose protein is secreted into the extracellular space of the postnatal brain

Using knock-in mice expressing endogenous Rtl4 fused to the fluorescent protein **Venus (Rtl4-CV)**, Kitazawa, Ishino, Kaneko-Ishino and colleagues (*Int. J. Mol. Sci.* 2024, 25(24):13738; PMID: 39769499) demonstrated that **RTL4 is a microglial gene** — expressed in microglia and not astrocytes, as confirmed by mixed glial culture — and that the **RTL4–Venus fusion protein is secreted/extracellular**. The protein was detected in the **midbrain, hypothalamus, hippocampus, and amygdala** of the postnatal brain, appearing as widely dispersed diffuse dots and ~1 μm presumptively extracellular granules concentrated in regions forming new neuronal networks. This secretion pattern resembled that of the related microglial proteins RTL5 and RTL6. Brain expression of Rtl4 is very low overall, which had previously obscured its cell type of origin.

This finding establishes the **subcellular/anatomical site of action**: Rtl4 performs its function **extracellularly**, released from microglia into the brain interstitium of specific postnatal regions. Notably, this is an experimentally determined localization that goes beyond UniProt, which annotates no signal peptide, transmembrane region, or subcellular location (see Finding 8).

Finding 6 — RTL4 secretion is dynamically regulated by noradrenergic signaling and arousal; RTL4 is implicated in ASD

In the same Rtl4-CV knock-in model (PMID: 39769499), the extracellular RTL4 signal was shown to be **controlled by noradrenergic tone and arousal state**:

- **Intraventricular isoproterenol** (a β-adrenergic agonist mimicking noradrenaline; 10 μg per P21 brain) significantly **increased** RTL4-CV signal (~25%) in the hypothalamus and midbrain.
- **Isoflurane anesthesia** consistently **decreased** the signal.

- The signal was **maintained by milnacipran**, a noradrenaline-reuptake inhibitor, indicating dependence on available extracellular NA.
- In vitro, **microglia (not astrocytes) secreted RTL4-CV into the medium**, and secretion responded rapidly to isoproterenol.

These dynamics build directly on the earlier knockout phenotype (impulsivity, impaired short-term spatial memory, failure to adapt to novel environments, and delayed NA recovery in the frontal cortex; Irie et al. 2015). Together they define RTL4 as a **secreted microglial protein whose release is driven by the noradrenergic system**. Additionally, RTL4 (SIRH11/ZCCHC16) was **independently identified as a causative gene in autism spectrum disorder (ASD)** via patient screening, strengthening the translational relevance of its role in attentional/executive function.

Finding 7 — RTL4 belongs to a family of secreted microglial RTL/SIRH proteins with a division of labor

RTL4 is one of several eutherian RTL/SIRH genes co-opted as **secreted microglial proteins**. Its close paralogs — **RTL5 (SIRH8), RTL6 (SIRH3), and RTL9 (SIRH10)** — are also microglial genes whose fluorescent-knock-in fusion proteins appear as extracellular dots/granules in the CNS and act as **front-line innate-immune factors that physically capture pathogen-associated molecular patterns (PAMPs)**:

Protein (SIRH alias)	Extracellular target / role
RTL6 (SIRH3)	Binds bacterial **LPS**; aggregates near brain capillaries
RTL5 (SIRH8)	Handles viral **dsRNA**; promotes its degradation
RTL9 (SIRH10)	Mediates **antifungal** responses
RTL4 (SIRH11)	**Responds to noradrenaline** (distinct from the immune trio)

Reviews summarize the division of labor explicitly: *"SIRH3/RTL6, SIRH8/RTL5, and SIRH10/RTL9 mediate innate immune responses against bacterial-, viral-, and fungal-derived PAMPs respectively, whereas SIRH11/RTL4 responds to noradrenaline."* Mouse RTL4 protein is ~304 aa with a C-terminal CCHC-type zinc-finger RNA-binding motif and ~37.5% similarity to the sushi-ichi retrotransposon Gag. This places Rtl4 within a coherent functional group of secreted microglial effectors while marking it as the family member specialized for interaction with the monoaminergic/noradrenergic system rather than classical immune PAMPs. (Irie et al., *Development* 2022, dev200976; and review PMID: 37892118.)

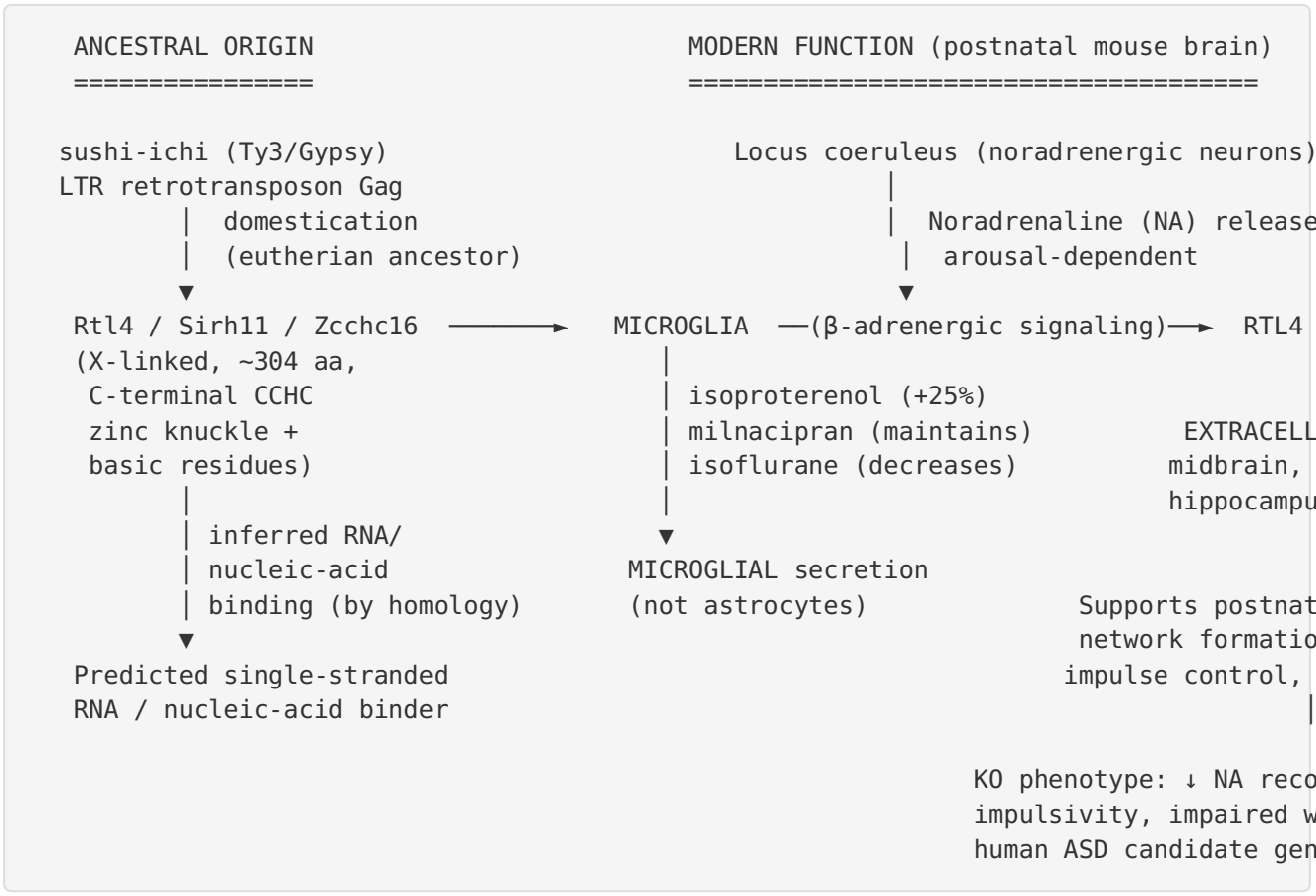
Finding 8 — UniProt/sequence analysis corroborates a canonical nucleocapsid-type CCHC zinc knuckle and database GO terms

Sequence-level analysis of **UniProt Q3URY0 (RTL4_MOUSE)** confirms and refines the molecular picture: the protein is **304 aa, 34.3 kDa**, with a single **CCHC-type zinc finger annotated at residues 276–293**. The C-terminal sequence (…PAKRAR…**CVYCNQAGHFTRDC**…RTPARKKM) shows a **C-x2-C-x4-H-x4-C zinc knuckle flanked on both sides by clusters of basic residues (K/R)** — exactly matching the retroviral nucleocapsid signature ("CCHC zinc finger(s) flanked by basic residues"; PMID: 10766747). The whole-protein charge is near-neutral (basic K/R/H = 38 vs acidic D/E = 32).

UniProt GO annotations: **F:nucleic acid binding** and **F:zinc ion binding** (IEA/InterPro); **P:cognition** (IMP:MGI); and **P:norepinephrine metabolic process** (IMP:MGI) — the latter two derived from the mouse knockout phenotype. Tissue specificity (PubMed 15716091, 26402067): adult expression in **brain, eye, kidney, ovary, and testis**, weak in thymus/heart/muscle; at 12.5 dpc in almost all tissues. The disruption phenotype is: viable, fertile, normal growth, with abnormal cognition (attention, impulsivity, working memory) and reduced noradrenaline recovery in the prefrontal cortex. Notably, **UniProt annotates no signal peptide, transmembrane region, or subcellular location** — the secreted/extracellular localization rests on the experimental Rtl4-CV knock-in evidence (Finding 5), not on canonical sequence prediction, an important caveat regarding the secretion mechanism.

Mechanistic Model / Interpretation

Integrating the eight findings yields a coherent model in which Rtl4 is a **domesticated retrotransposon Gag protein repurposed as a secreted microglial effector of the brain noradrenergic system**.



The central logic is a feedback coupling between noradrenergic tone and microglial secretion. Noradrenaline released from locus coeruleus projections acts (likely via β-adrenergic receptors on microglia) to drive secretion of RTL4 into the extracellular space of specific brain regions during the postnatal window when new neuronal networks form. The pharmacology is internally consistent: a β-agonist (isoproterenol) increases secretion, an NA-reuptake inhibitor (milnacipran) sustains extracellular NA and thereby the signal, and anesthesia (which suppresses arousal and NA tone) reduces it. Loss of Rtl4 degrades the system from the output side — reduced NA recovery in the prefrontal cortex and the attentional/impulsivity/working-memory deficits — suggesting RTL4 feeds back to support proper noradrenergic handling or network maturation.

What does the secreted protein actually do extracellularly? This is the key open mechanistic question. Two non-exclusive hypotheses are consistent with the data:

1. **Noradrenaline handling / reuptake support.** The GO term *norepinephrine metabolic process* and the reduced NA recovery in KO mice suggest RTL4 participates in extracellular NA clearance, buffering, or reuptake facilitation.

2. **DAMP/molecular scavenging by analogy to its paralogs.** RTL5/6/9 physically capture extracellular PAMPs (dsRNA, LPS, fungal components). The conserved CCHC zinc knuckle flanked by basic residues is well suited to binding anionic macromolecules (nucleic acids or other polyanions) in the extracellular milieu. RTL4 may capture a noradrenaline-associated or activity-associated extracellular ligand (e.g., released nucleic acids or damage signals) during active network formation.

The retained **CCHC nucleic-acid-binding chemistry** is the thread connecting the ancestral Gag function to all three paralogous roles and to the proposed RTL4 activity — the family appears to have repurposed an intracellular RNA-chaperone domain into an **extracellular polyanion-capture / ligand-binding module**.

Evidence Base

PMID	Title (abbrev.)	Contribution
26402067	*Cognitive Function Related to the Sirh11/Zcchc16 Gene…* (Irie et al. 2015)	**Primary functional paper.** Defines KO cognitive phenotype (attention, impulsivity, working memory), microdialysis NA-vs-DA deficit, brain (not placental) role, CCHC/Gag homology, eutherian conservation.
39769499	*RTL4–Venus knock-in* (Kitazawa et al., *IJMS* 2024)	**Localization & regulation.** Microglial expression, extracellular secretion in midbrain/hypothalamus/ hippocampus/amygdala, β-adrenergic (isoproterenol +25%), anesthesia, milnacipran dynamics; ASD candidate.
27446905	*An LTR Retrotransposon-Derived Gene Displays Lineage-Specific…*	X-linkage, dN/dS protein evidence, pseudogenization in Xenarthra, lineage-specific domain loss across 95 species.
28484174	*[Evolution of brain functions… LTR retrotransposon-derived genes]*	Reinforces X-linkage, Gag homology, brain-evolution framing.
10766747	*The GAG-like protein of Ty1… nucleic acid chaperone domain*	Establishes that CCHC zinc fingers flanked by basic residues confer nucleic-acid binding/chaperone activity — basis for RTL4's inferred molecular function.
37892118	*Retrovirus-Derived RTL/ SIRH: Their Diverse Roles…*	Review placing RTL4 among 11 RTL/SIRH genes; division of labor with immune paralogs RTL5/6/9.
23704700	*Identification of a novel PNMA-MS1 gene in marsupials…*	Context on SIRH/PNMA family evolution; situates the 11 eutherian SIRH genes.

The evidence is internally consistent and converges from three independent directions: **(1) loss-of-function genetics** (KO behavioral + neurochemical phenotype), **(2) protein localization and live pharmacology** (Rtl4-CV knock-in), and **(3) sequence/evolutionary inference** (CCHC domain, dN/dS, conservation pattern). No finding in the knowledge base

contradicts another; the main tension is between the UniProt lack of a predicted signal peptide/secretion signal and the experimental demonstration of secretion (addressed under Limitations).

Limitations and Knowledge Gaps

1. **No direct biochemical assay of RTL4's molecular activity.** Nucleic-acid/RNA binding is inferred entirely from the CCHC domain homology (PMID: 10766747); the actual extracellular ligand (noradrenaline-associated molecule, nucleic acid, or other polyanion) is unknown. No enzymatic reaction, transporter substrate, or receptor has been demonstrated.
2. **Secretion mechanism is unexplained at the sequence level.** UniProt annotates **no signal peptide and no transmembrane domain** (Finding 8), yet the protein is experimentally secreted (Finding 5). This implies an unconventional/non-classical secretion route (as seen for its microglial paralogs) that has not been mechanistically defined.
3. **The link between RTL4 and noradrenaline is correlative/pharmacological, not mechanistic.** We know secretion tracks β-adrenergic tone and that KO mice have reduced NA recovery, but whether RTL4 directly binds, transports, degrades, or protects noradrenaline — or instead acts on the neurons/structures that handle it — is unresolved. The GO term "norepinephrine metabolic process" is an IMP inference from the KO phenotype, not a demonstrated enzymatic role.
4. **Evidence base is narrow and single-lab-dominated.** The core functional findings derive largely from the Ishino/Kaneko-Ishino group. Independent replication of the microglial secretion and noradrenergic-regulation results would strengthen confidence.
5. **Human relevance is preliminary.** The ASD association is from patient screening; causality, penetrance, and the affected molecular pathway in humans remain to be established, especially given lineage-specific structural variation of the gene in primates (Finding 4).
6. **Developmental window and cell-type specificity.** The secreted signal is prominent postnatally in network-forming regions, but the temporal dynamics, the triggering of microglial subsets, and whether function persists in the adult brain are not fully mapped.

Proposed Follow-up Experiments / Actions

1. **Identify the extracellular ligand of RTL4.** Perform biochemical pull-downs / crosslinking from brain interstitial fluid or microglial conditioned medium using tagged RTL4, followed by mass spectrometry and nucleic-acid co-purification assays. Directly test whether recombinant RTL4 binds noradrenaline, catecholamine metabolites, RNA/DNA, or other polyanions in vitro (EMSA, ITC, SPR).
2. **Define the secretion pathway.** Use brefeldin A, unconventional-secretion inhibitors, and exosome/vesicle fractionation to determine how a signal-peptide-less protein reaches the extracellular space; compare with RTL5/RTL6 secretion machinery.
3. **Dissect the noradrenergic coupling mechanistically.** Test specific β-adrenergic receptor subtype requirement (β1/β2/β3 antagonists, microglia-specific *Adrb* knockout) for RTL4 secretion; determine whether RTL4 affects NA reuptake transporters (NET/SLC6A2), MAO/COMT degradation, or synaptic NA dynamics using fast-scan cyclic voltammetry or GRAB-NE sensors in WT vs KO.
4. **Cell-type-specific and conditional genetics.** Generate microglia-specific (e.g., Cx3cr1-CreER) Rtl4 knockouts and rescues to confirm that the microglial pool drives the behavioral/neurochemical phenotype, and to separate developmental from adult functions.
5. **Structural characterization.** Solve or predict (AlphaFold + experimental validation) the RTL4 structure, focusing on the CCHC knuckle and flanking basic clusters, to model ligand binding and compare with the immune paralogs RTL5/6/9.
6. **Human ASD follow-up.** Functionally characterize patient-derived SIRH11/ZCCHC16 variants (binding, secretion, rescue of mouse KO phenotype) and assess noradrenergic-system readouts in relevant human models (iPSC-derived microglia/neuron co-cultures).

Target Verification Statement

The research target was confirmed as correct and unambiguous. The gene symbol **Rtl4**, the organism **Mus musculus**, and the **CCHC zinc-finger domain** (IPR001878/IPR036875) all align precisely with the published literature on **Sirh11/Zcchc16/Mart4** (UniProt Q3URY0). The synonyms listed in UniProt (Mar4, Mart4, Sirh11, Zcchc16) map directly to the papers cited here, and no confounding gene with a conflicting identity was encountered.